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A comprehensive review on brown spot disease of rice: etiology, epidemiology, management strategies and future directions

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Abstract

Rice (Oryza sativa L.) is the second largest staple grain among cereal crops, feeding more than half of the worlds' population. Many fungal diseases damage rice crops, leading to considerable yield loss. Bipolaris oryzae, the teleomorph Cochliobolus miyabeanus, is the cause of brown spots on rice, a global problem known to significantly reduce grain production up to 52 % quantitatively and qualitatively. Under conditions of direct seeding, drought and low input management, the brown spot disease is most significant. The disease is also historically important as it caused a disastrous outbreak in the Bengal Province that culminated in the Great Bengal Famine (1943), which left 2.1 to 3 million people starved to death. The brown spot remains terrible when considering the current scenario for rice deterioration. A broad host range, pathogenicity and molecular diversity characterize the pathogen. In this present article, we have emphasized the epidemiology, the prevention techniques that are currently in use and several quantitative and qualitative gaps regarding disease management that, if filled, would have a significant impact on crop disease control and the long-term sustainability of rice and are relevant to farmers' current circumstances.

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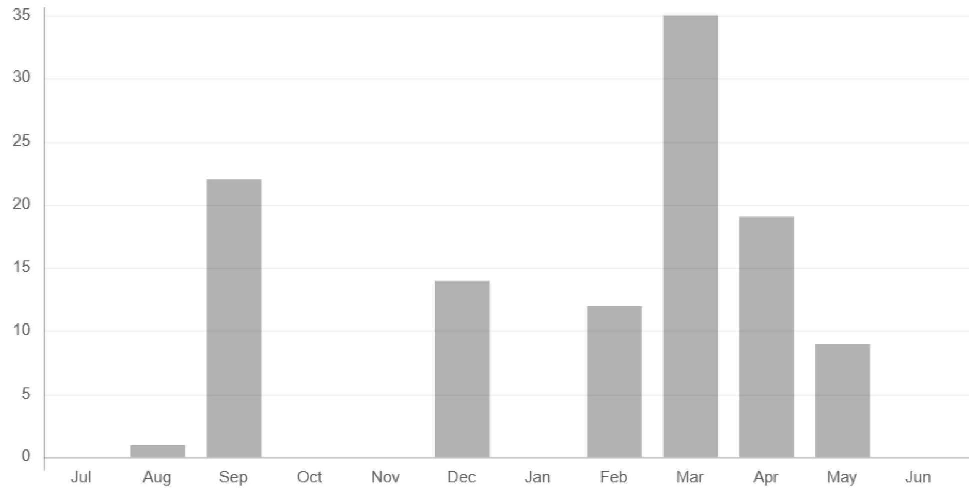


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REVIEW ARTICLE

A comprehensive review on brown spot disease of rice: Etiology, epidemiology, management strategies and future directions

Tenzing Phurbu Bhutia[1], Yenda Damodhara Rao[1], Priyanka Kharel[1], Prayashma Rai[1], Meenakshi Rana[1] & Seweta Srivastava[1*]

Abstract

Rice (*Oryza sativa* L.) is the second largest staple grain among cereal crops, feeding more than half of the worlds' population. Many fungal diseases damage rice crops, leading to considerable yield loss. *Bipolaris oryzae*, the teleomorph *Cochliobolus miyabeanus*, is the cause of brown spots on rice, a global problem known to significantly reduce grain production up to 52 % quantitatively and qualitatively. Under conditions of direct seeding, drought and low input management, the brown spot disease is most significant. The disease is also historically important as it caused a disastrous outbreak in the Bengal Province that culminated in the Great Bengal Famine (1943), which left 2.1 to 3 million people starved to death. The brown spot remains terrible when considering the current scenario for rice deterioration. A broad host range, pathogenicity and molecular diversity characterize the pathogen. In this present article, we have emphasized the epidemiology, the prevention techniques that are currently in use and several quantitative and qualitative gaps regarding disease management that, if filled, would have a significant impact on crop disease control and the long-term sustainability of rice and are relevant to farmers' current circumstances.

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Keywords

- Bipolaris oryzae
- brown spot
- epidemiology
- etiology
- management
- rice

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